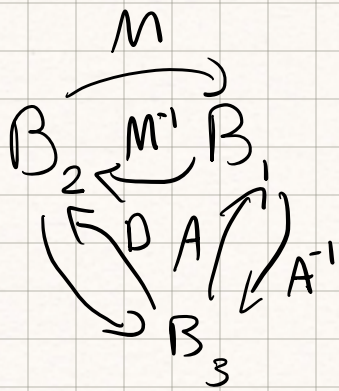




$$M = \begin{bmatrix} 1 & -1 \\ -2 & -1 \end{bmatrix}$$

$$M^{-1} = \begin{bmatrix} 1/3 & -1/3 \\ 2/3 & -1/3 \end{bmatrix}$$

$$A = \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 3/5 & -2/5 \\ -2/5 & 3/5 \end{bmatrix}$$

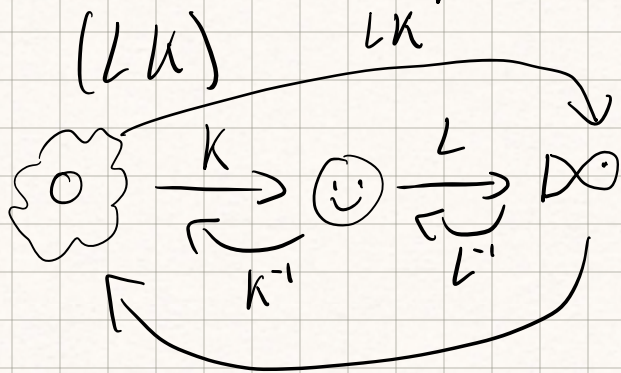
$$D = \begin{bmatrix} 1/3 & -1/3 \\ -8/3 & -7/3 \end{bmatrix}$$

$$D^{-1} =$$

$$D = M^{-1}A$$

$$\begin{bmatrix} 1/3 & -1/3 \\ 2/3 & -1/3 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 1/3 & -1/3 \\ -8/3 & -7/3 \end{bmatrix}$$

inverse of a product



$$K^{-1}L^{-1} = (LK)^{-1}$$

$$D = A^{-1}M = \begin{bmatrix} 3/5 & -2/5 \\ -2/5 & 3/5 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -2 & -1 \end{bmatrix} = \begin{bmatrix} 7/5 & -1/5 \\ -8/5 & -1/5 \end{bmatrix}$$

$$D = \frac{1}{\underbrace{\frac{-7}{9} - \frac{-8}{9}}_{-\frac{3}{5}}} \begin{bmatrix} -7/3 & 1/3 \\ 8/3 & 1/3 \end{bmatrix} =$$

Lem: if $B_1 = [v_1, v_2]$ and $B_2 = [w_1, w_2]$

are 2 bases for V and $w_1 = av_1 + bv_2$

$$w_2 = cv_1 + dv_2$$

then the linear function taking B_2 coords to B_1 coords is

$$\begin{bmatrix} a & c \\ b & d \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix}_{B_3} = \begin{bmatrix} \quad \\ \quad \end{bmatrix}_{B_2} = \mathbb{O} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1/3 & -1/3 \\ -8/3 & -7/3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} -1/3 \\ -22/3 \end{bmatrix}_{B_2}$$

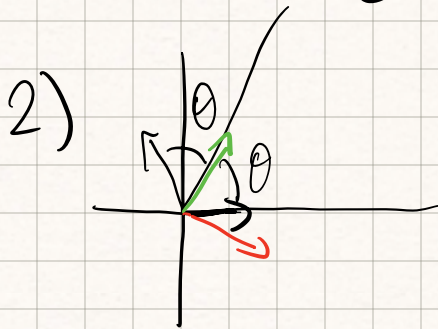
Compute the matrix for

1) dilate by $\frac{1}{2}$ — $\begin{bmatrix} 1/2 & 0 \\ 0 & 1/2 \end{bmatrix}$

2) reflect over $y=2x$

3) rotate by $\pi/3$ $\begin{bmatrix} 1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & 1/2 \end{bmatrix}$

in standard basis



$$\begin{bmatrix} 1 \\ 2 \end{bmatrix}_{v_1} \lambda_1 \quad \begin{bmatrix} 2 \\ -1 \end{bmatrix}_{v_2} (-1) \lambda_2$$

the matrix for 2) in eigenvalue basis is $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

Change from eigen to standard basis $\leftarrow$ Change from standard to eigen basis

$$\begin{bmatrix} 1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & 1/2 \end{bmatrix} \begin{bmatrix} \rightarrow \\ \rightarrow \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} \rightarrow \\ \rightarrow \end{bmatrix} \begin{bmatrix} 1/2 & 0 \\ 0 & 1/2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix} = \Phi$$

$\rightarrow$ eigenvector basis written in terms of standard basis

Calculate $[2]$ in standard basis

$$\Phi^{-1} = \frac{1}{-1-4} \begin{bmatrix} -1 & -2 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} 1/5 & 2/5 \\ 2/5 & -1/5 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1/5 & 2/5 \\ 2/5 & -1/5 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 1/5 & 2/5 \\ -2/5 & 1/5 \end{bmatrix} = \begin{bmatrix} -3/5 & 4/5 \\ 4/5 & 3/5 \end{bmatrix}$$